

DataFlippers : A drag and drop interface to build machine learning models

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Software

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Statement of need

A lack of programming skills limits machine learning (ML) education ([Bart et al., 2016](#); [Brunner & Kim, 2016](#); [Thayer, 2020](#)). This is because many ML tools require coding. This lack of accessibility limits widespread ML literacy, hindering critical evaluation of AI in modern life ([Jain, 2021](#); [Provost & Fawcett, 2013](#)). To address this, we need intuitive tools that prioritize a low barrier to entry. DataFlippers aims to provide high schoolers and early undergraduates a GUI based software to explore core data science and build ML models without requiring prior programming experience.

State of the field

There are several no-code/low code platforms available on the internet. However, these are often designed with power users in mind, making them frustrating to novices. Additionally these softwares are often geared toward commercial data science use, with a focus on integration with common business tools.

Name	Description	Target Audience	License	Drag and drop
DataBricks	Generative AI	Businesses	Paid / Commercial	No
Power BI	Visualization Interface	Businesses	Paid / Commercial	Yes
Rapid Miner	Training / Visualization	Data Scientists	Free for individuals	No
JASP	Statistics / Visualization	Upper Undergraduate / Graduate	Free	No

The platforms that are free and tailored to students, such as JASP ([JASP Team, 2025](#)), are catered to teaching students statistics, Rather than expressly teaching them ML.

20 Software design

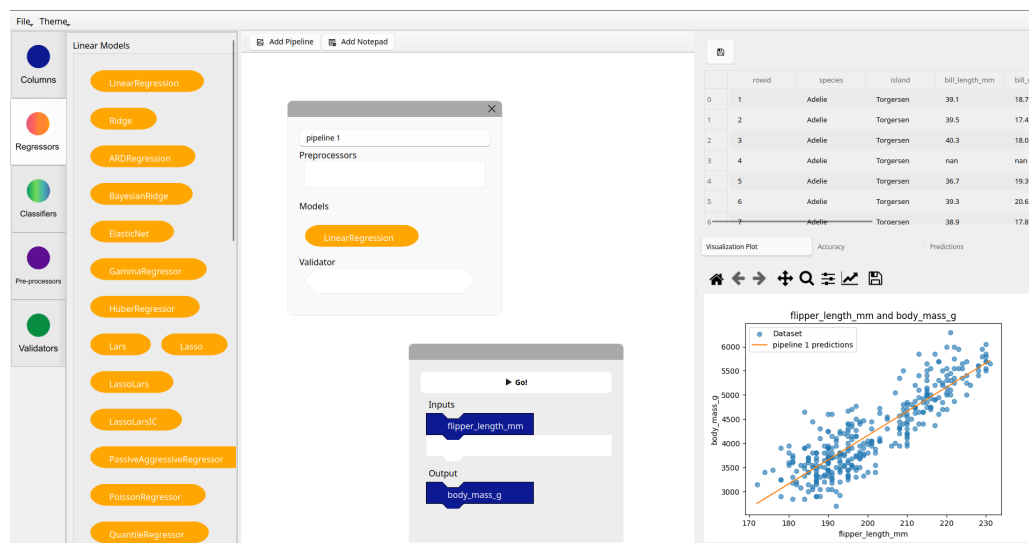


Figure 1: Image of the fullscreen DataFlippers project page.

21 Inspiration

Scratch(Resnick et al., 2009), a visual programming language designed for children, offers a compelling model for accessible computational learning. Its intuitive drag-and-drop interface allows beginners to grasp fundamental programming concepts without needing to decipher complex syntax. Scratch's interface has been proven to be effective at teaching novices programming concepts(Armoni et al., 2015). This drag and drop model provided inspiration for DataFlippers. The column block shape is modeled after the Scratch 'stack' block, the model blocks after Scratch's 'reporter' blocks, and the validators after Scratch's 'boolean' blocks.

30 Software architecture and core project libraries

The language for this software is Python and PyQt, this is because python possesses several commonly used ML libraries(Burridge & Fekete, 2022), such as pandas(team, 2020), matplotlib(Hunter, 2007), and scikit-learn(Pedregosa et al., 2011).The core ML models are provided by scikit-learn(Pedregosa et al., 2011). There are several reasons for choosing scikit-learn, one of them is portability. As scikit-learn is very common within the entry level ML field, the skills that novices learn from DataFlippers could translate easily to programmatic skills, should the novice decides to learn programming. Additionally, scikit-learn is very friendly for first time users, featuring extensive documentation, which can be viewed by hovering on each items tooltip. pandas(team, 2020) is used for handling the data. This is because pandas supports the importing of multiple standard file type formats, such as CSV, Excel, and Parquet. Matplotlib is utilized for the plotting features, this is because matplotlib interfaces well with the PyQt background, as well as being another standard ML library.

44 Drag and drop design

The drag and drop interface is designed to reduce the syntactic complexity of programming down to drag and drop blocks, with the layout and design imitating underlying Python libraries. This ensures that users have an low floor to learning. Each model, preprocessor,

48 and validator corresponds to a sklearn class. Each draggable block has a popup menu that
49 is available on click. Clicking on a column opens a small descriptive statistics panel, while
50 clicking on other blocks opens a hyperparameter tuning menu.

51 Users are also enabled to perform basic statistical analysis. This can be done by removing
52 all of the pipeline blocks, plotting via the “Inputs and Outputs” block. This can allow for
53 the user to perform various statistical analysis tasks.

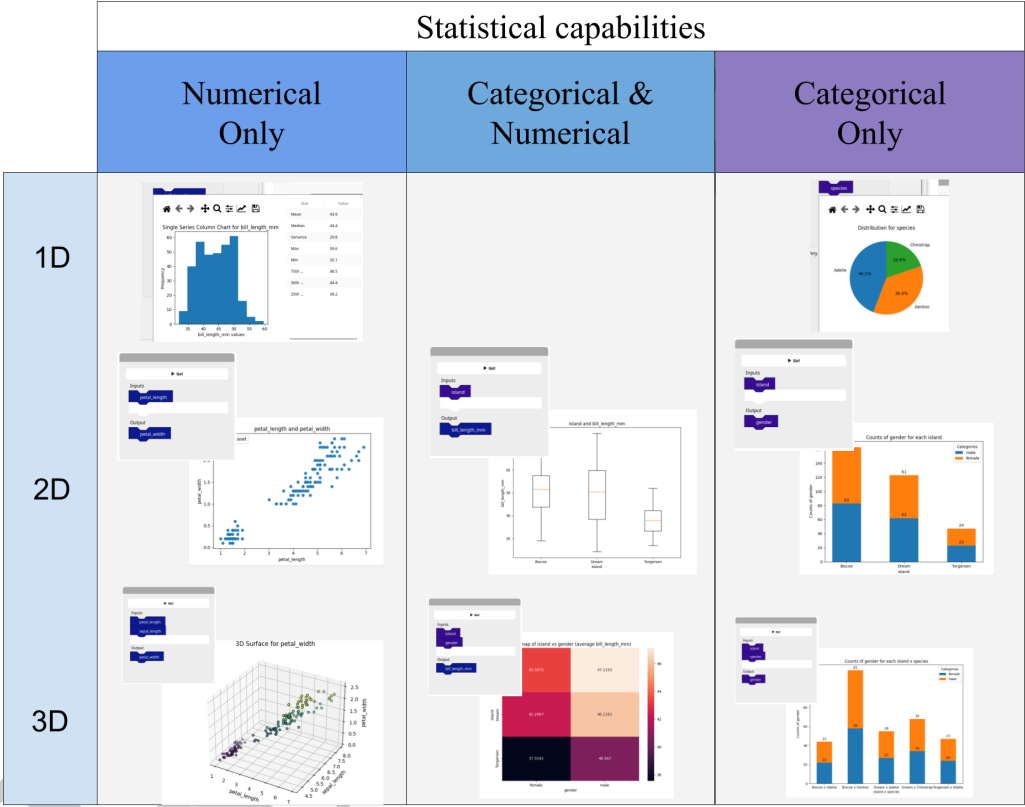


Figure 2: The descriptive statistical abilities of the dataflippers software. Each of these can be made entirely from drag and drop.

54 **Teaching materials**

55 **Informal user testing**

56 DataFlippers has undergone multiple informal usability testing with a small groups of
57 novice users. While using an early version of DataFlippers in October of 2025, a user
58 remarked his disappointment at being unable to plot multiple models at the same time.
59 Feedback from these sessions was used to improve the overall design of the drag and drop
60 plotting, and later add a multi-model setup.

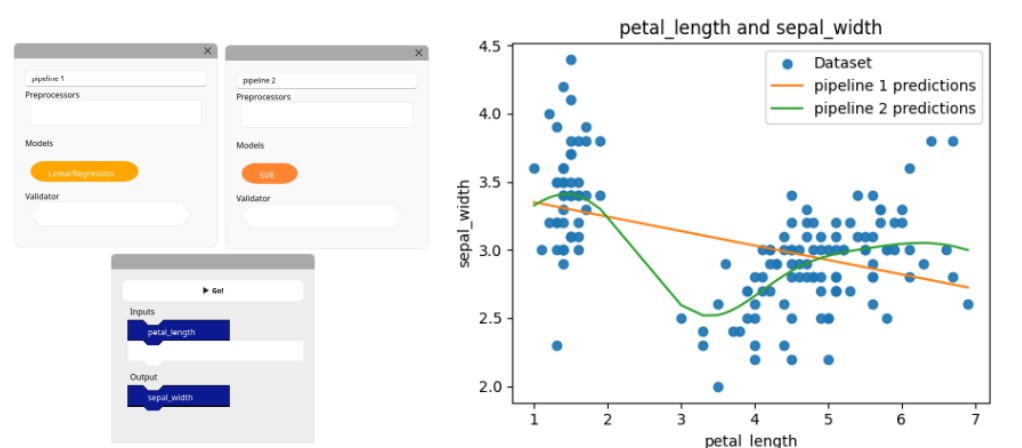


Figure 3: Image showing the model comparison. This would allow novices to understand and visually see the differences between different types of ML models.

DataFlippers later underwent a second usability test. In which users were be instructed to complete common data science and ML tasks. The “penguins” dataset consistently elicited the most positive feedback from participants, with participants stating that the penguins offered easily visualizable attributes such as flipper length, weight, and species. This provided inspiration for both the name and the logo for DataFlippers.

User documentation

The main DataFlippers page features a [walkthrough tutorial](#) on how to use the software. The website also features descriptions of core ML concepts such as [classification](#), [regression](#), [statistical analysis](#), and [pre-processors](#). The source code is available at [DataFlippers GitHub](#). The software can be downloaded on windows or linux for free at the [DataFlippers homepage](#)

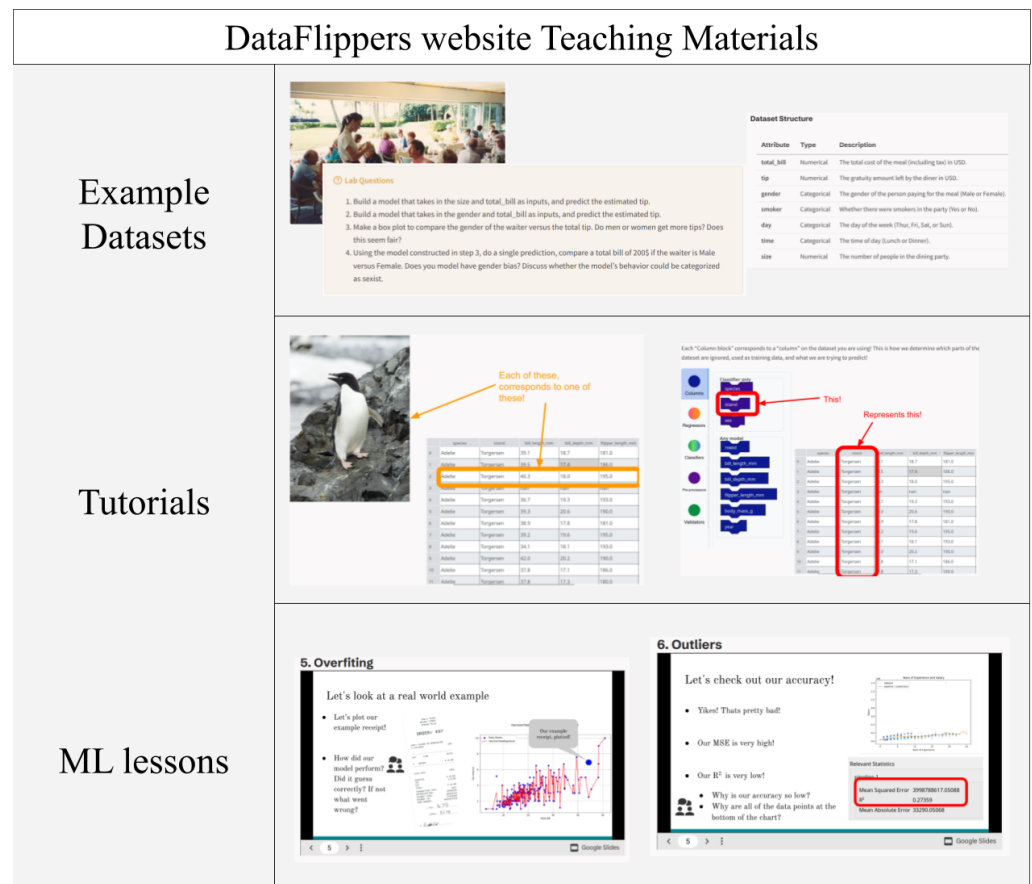


Figure 4: Image showing the different types of educational materials available on the DataFlippers homepage.

Example datasets and Lab questions

DataScratch comes pre-loaded with several example datasets. These example datasets enable novices to start learning right away, without having to procure a dataset first. These datasets are a blend of commonly used data science teaching datasets such as [palmer's penguins](#) (Horst et al., 2020), [diamonds dataset](#) (Waskom, 2021) and the [iris dataset](#) (Fisher, 1936). As well as datasets that would be enticing to a younger audience, such as a dataset containing information about [pokemon](#) (Banik, 2018) and a dataset about [minecraft biome statistics](#) (Pillot, 2026). These example datasets come with 3-5 lab questions each, which can be accessed from the main DataFlippers website. These lab questions are intended to be part of an in-person class exercise, where students could participate in groups of 2-3 to complete the lab questions, with guidance from an educator.

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